Using Unsupervised Machine Learning for Galaxy Clustering

Team 3 - Machine Learning Project

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May 28, 2026

The big picture

- Classifying galaxies is easy

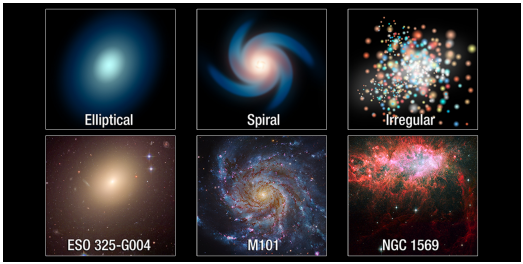


Figure: Low redshift galaxy classification

- Just look at their shape, and this will tell you their physical properties...
- ... Right?

The big picture

- What about now?



Figure: High redshift galaxy "classification"

- The further away a galaxy is, the older it is, and the more irregular it looks
- Is there a way to classify these high redshift galaxies such that they match their physical properties?

The big picture

What we know:

- The familiar **Hubble sequence** (Elliptical / Spiral / Irregular) was built on *low*-redshift, nearby galaxies.
- This dataset was constructed using humans (Galaxy Zoo)
- At $z > 2$, galaxies look qualitatively different: clumpier, more disturbed, less symmetric – the labels we have don't cleanly apply.
- Question: can **unsupervised learning** discover meaningful morphology classes that directly correlate with physical properties?

The paper we replicate

Tohill et al. 2023 – arXiv:2306.17225

"Unsupervised Galaxy Morphological Classification at High-z with JWST/CEERS"

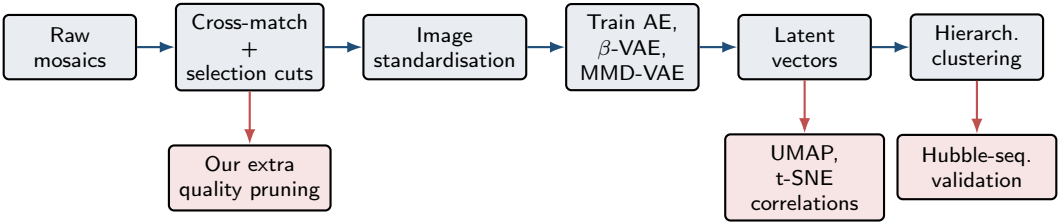
- **Goal of the paper:** train a **Maximum Mean Discrepancy VAE** (MMD-VAE) on JWST/CEERS images, then **hierarchically cluster** the latent vectors to discover morphology groups.

Our objectives – what we set out to do

What the paper does *not* do (and we will):

1. **Replicate** the Tohill et al. pipeline end-to-end, from raw JWST mosaics to clustered latents.
2. **Justify the paper's design choices** by comparing alternative models (plain AE, β-VAE) and alternative clustering / dimensionality-reduction methods (UMAP, t-SNE, K-means).
3. **Provide a measure of "good clustering"** that was missing in the paper, by applying the same pipeline to the known **Hubble sequence** classification and verifying its performance.

Pipeline overview



Blue boxes = paper replication. Red boxes = our contributions.

Datasets at a glance

CEERS (JWST/NIRCam, primary)

- Cox et al. 2025 DR1.0 photometric catalog ($\sim 80{,}000$ sources).
- NIRCam mosaics in 7 filters (F115W ... F444W).
- Image scale: $0.03''$/pix.

CANDELS (HST + catalog)

- Santini et al. 2015 mass catalog: ID, RA/Dec, $z_{\rm best}$, $M_\star$, AGN flag.
- Used to attach physical labels via positional cross-match.

COSMOS-Web 2025 (JWST, extension)

- Shuntov et al. 2025 DR1 master catalog.
- Different patch of sky – tests out-of-distribution behaviour.
- Used as a proxy ground truth for the Hubble-sequence validation.

Part 1 – Pre-processing pipeline

Speaker: Antonio

Steps:

1. **Cross-match** CEERS $\leftrightarrow$ CANDELS within $1''$.
2. Apply paper selection cuts:
   - $z > 2$
   - $\log_{10}(M_\star/M_\odot) > 8.5$
   - **No Active Galactic Nuclei**
   - Optional CEERS quality flags (`STAR_FLAG`, `BAD_REGION_FLAG`).

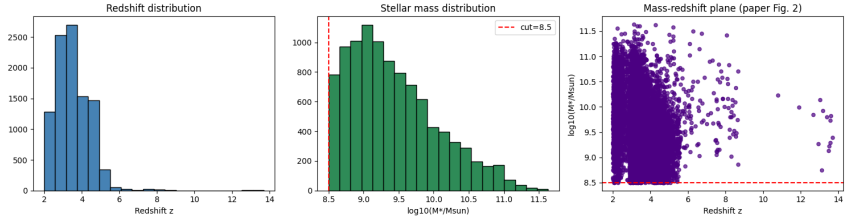


Figure: Results after prunning

Part 1 – Pre-processing pipeline

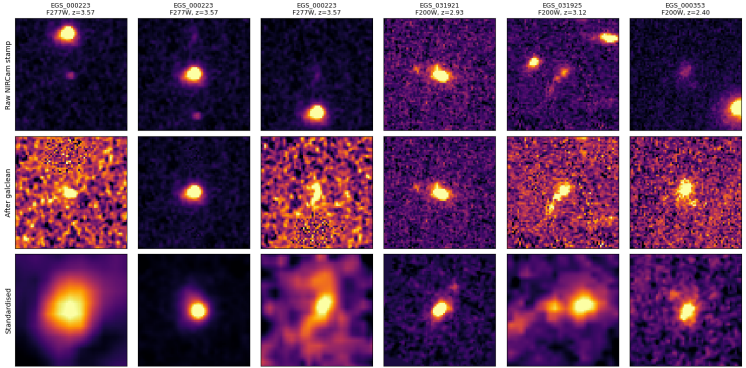
Speaker: Antonio

3. For each surviving galaxy:
   - Pick the NIRCam filter closest to **rest-frame V-band** $\lambda = 0.55\,(1+z)\,\mu$m ("normalising" redshift).
   - Download *only* the mosaics we need ($\sim$1–3 GB each).
   - Cut a 128×128 pixel stamp around each galaxy.
4. Standardise the stamp.
   - **Galclean algorithm** (de Albernaz Ferreira & Ferrari 2018):
     1. Detect all sources
     2. Find the central one
     3. Mask the others
     4. Refill the mask and empty space with random draws from $\mathcal{N}(\mu_{\rm bg}, \sigma_{\rm bg})$ of the background.
   - Measure the **position angle (PA)**, **axis ratio** and **Petrosian radius** (R_P) of the central source.
   - Rotate so the major axis is horizontal.
   - Rescale so that $R_P = 15$ pixels (target). Only upscale; never downscale (would lose

Part 1 – Pre-processing pipeline

Speaker: Antonio

- Rotate so the major axis is horizontal.
- Rescale so that $R_P = 15$ pixels (target). Only upscale; never downscale (would lose pixel information).
- Crop to a fixed 64×64 pixel stamp.



NOTES – Part 1 – Pre-processing pipeline

Speaker: Antonio

Filter assignment: for each galaxy we pick the NIRCam filter whose pivot wavelength is closest to $0.55 \times (1 + z)\,\mu$m, so all galaxies are imaged in their own **rest-frame V-band** – as if we were standing next to them. This reproduces Table 1 of the paper.
Why preprocessing matters here: the autoencoder is not given any morphology label; it can only "see" what we put in front of it. Any artefact (rotation, scale, background, neighbouring galaxy) becomes a fictitious axis in the latent space.

Bonus: non-parametric morphology

Speaker: Antonio

We also compute the **CAS statistics** on every cleaned stamp – not used for training but kept as auxiliary features to interpret clusters later.

Concentration

$$C = 5 \log_{10}(R_{80}/R_{20})$$

Ratio of radii enclosing 80% and 20% of the total light. High $C \Rightarrow$ centrally concentrated (early-type, bulge).

Asymmetry

$$A = \frac{\sum |I - I_{180}|}{\sum |I|} - A_{\mathrm{bg}}$$

How much light remains after

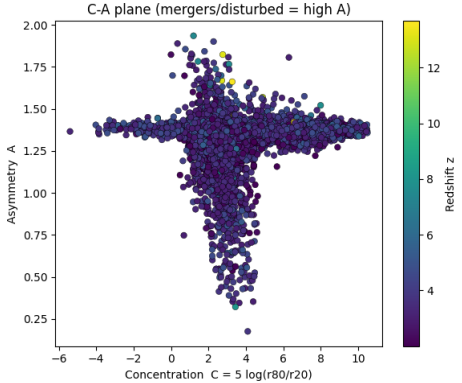


Figure: Contentration vs Assymetry plot

Final dataset assembly

- Per-galaxy record: `Object_ID`, `RA`, `Dec`, `Redshift_Z`, `Log_Stellar_Mass`, `Image_Filter`, `Image`, `Image_Cleaned`, `Image_Standardised`, `Position_Angle`, `Axis_Ratio`, `Petrosian_Radius`, `Asymmetry`, `Concentration`, `...`
- Saved as a pickle (full, with image arrays) and CSV (tabular only).
- After paper cuts: roughly $6{,}800$ entries – close to the paper's reported $6{,}869$.
- This becomes the input to the model notebook.

Architecture: The Autoencoder

Shared encoder/decoder architecture used by AE, β-VAE and MMD-VAE

Encoder — squeezes $(1, 64, 64)$ into $z \in \mathbb{R}^{23}$

- Convolution stack — three layers, each halving the spatial size:

$$(1, 64, 64) \;\to\; (32, 32, 32) \;\to\; (64, 16, 16) \;\to\; (128, 8, 8)$$

- Each step: Conv2D (kernel 4, stride 2, pad 1), BatchNorm, LeakyReLU(0.2)
- Bottleneck: Flatten $\to$ Dense(256) $\to$ two heads for μ and $\log \sigma^2$
- Reparameterisation: $z = \mu + \varepsilon \cdot \sigma, \quad \varepsilon \sim \mathcal{N}(0, 1)$

$\hookrightarrow$ AE uses μ directly (no sampling); VAE variants sample z

Decoder — rebuilds $(1, 64, 64)$ from $z \in \mathbb{R}^{23}$

- Projection: Dense(256) $\to$ Dense(8192) $\to$ reshape to $(128, 8, 8)$
- Transposed conv stack — three layers, each doubling the spatial size:

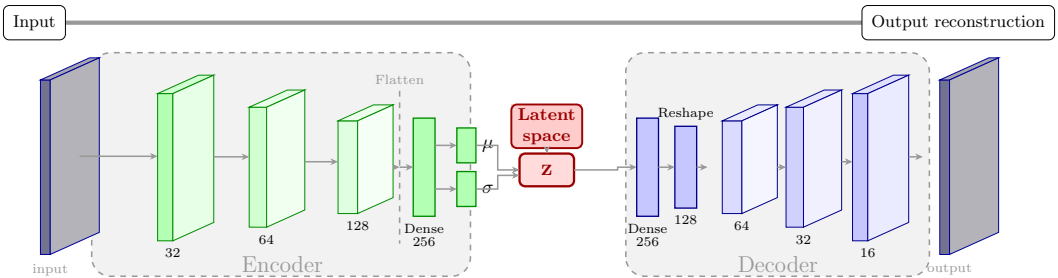
$$(128, 8, 8) \;\to\; (64, 16, 16) \;\to\; (32, 32, 32) \;\to\; (1, 64, 64)$$

- Each step: ConvTranspose2D (kernel 4, stride 2, pad 1), BatchNorm, LeakyReLU(0.2)
- Output activation: Sigmoid $\in [0, 1]$ to match normalised inputs

Architecture: The Autoencoder

Shared encoder/decoder architecture used by AE, β-VAE and MMD-VAE

This visualization was insipred by Tohill et al.



β-VAE and MMD-VAE Loss Functions

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}$$

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(\hat{x}|z) \right]$$

$$\begin{aligned} \mathcal{L}_{\text{KL}} &= D_{\text{KL}} \left[q(z|x) \,\|\, p(z) \right] \\ &= \frac{1}{2} \sum_j \left(1 + (\log \sigma_j)^2 - \mu_j^2 - \sigma_j^2 \right) \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{\text{MMD}} = \mathbb{E}_{p(z),p(z')} \left[k(z, z') \right] + \mathbb{E}_{q(z),q(z')} \left[k(z, z') \right] \\ - 2\mathbb{E}_{p(z),q(z')} \left[k(z, z') \right] \end{aligned}$$

$$\mathcal{L}_{\text{MMD-VAE}} = \lambda \cdot \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{MMD}}$$

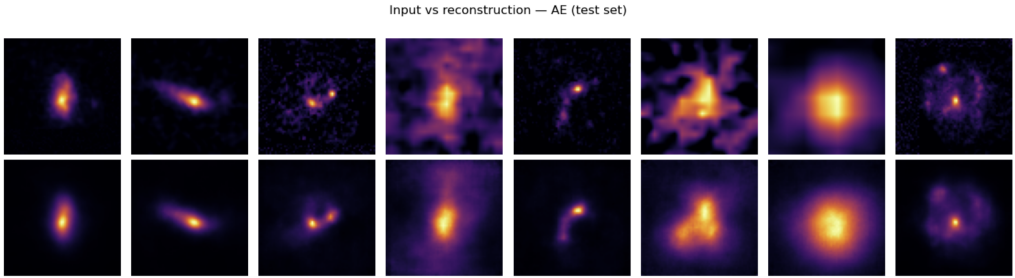
Training, Hyperparameter Tuning, Testing, and Outputs

1. Train / test / validate split: 80 / 10 / 10.
2. Training:
   1. Adam optimizer,
   2. Learning rate halving on plateau (patience = 20),
   3. Early stopping (patience = 50) in the training loop.
3. Hyperparameter tuning: conducted on a subset of the data for speed.
   1. $\lambda \in \{1, 10, 100, 1000\}$ for the MMD-VAE
   2. $\beta \in \{0.0001, 0.01, 0.1, 0.5, 1.0\}$ for the β-VAE.
4. Evaluation: reconstruction loss and aperture loss, visual comparisons of input galaxies and their reconstructions.
5. Output: Saved the latent vectors obtained from all three models to be used in the clustering.

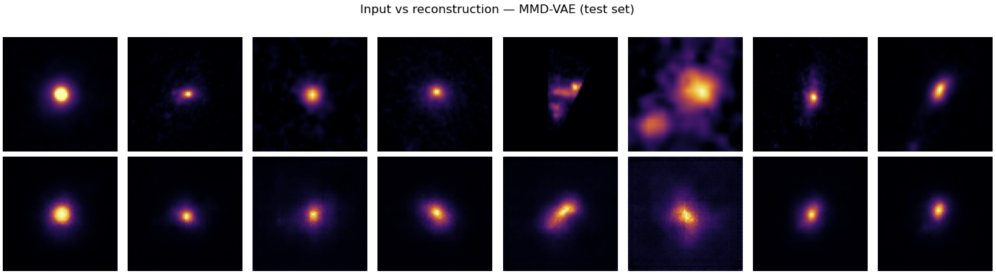
Loss metrics: an example ($\lambda = 100, \beta = 0.01$)

Model	Total loss	Recon	Reg term	Aper
MMD-VAE	0.3293	0.0002	0.3115	0.0236
β-VAE	0.0006	0.0005	0.0165	0.0732
AE	0.0001	0.0001	—	0.0052

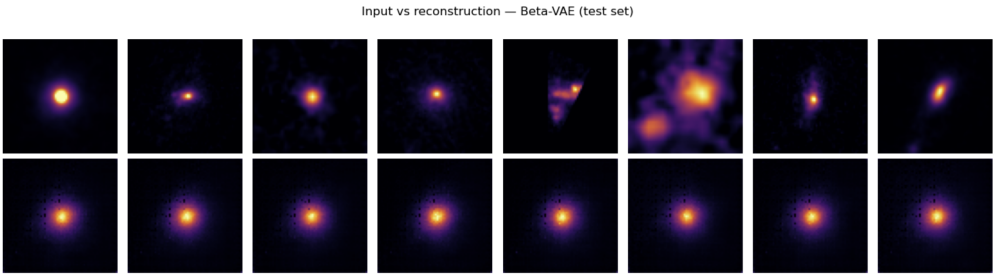
Reconstructions: Autoencoder



Reconstructions: MMD-VAE with $\lambda = 100$

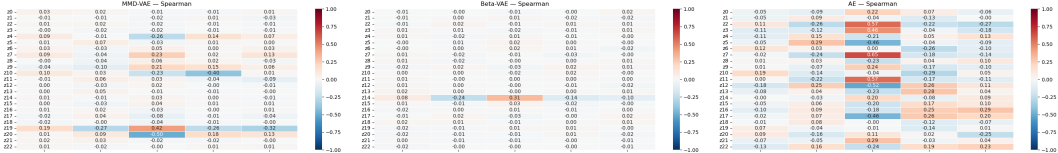


Reconstructions: β-VAE with $\beta = 0.5$



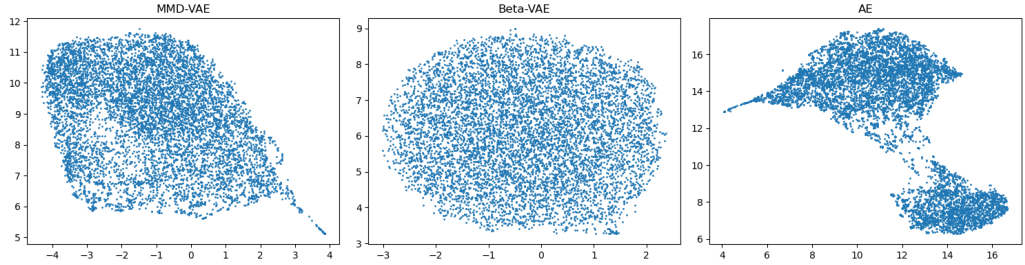
Analyzing the latent space

Spearman correlation of latent dimensions to physical parameters



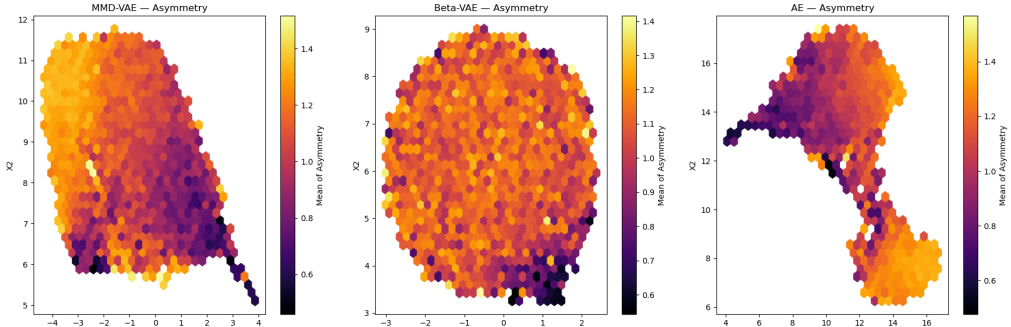
Analyzing the latent space

Dimensionality reduction with UMAP



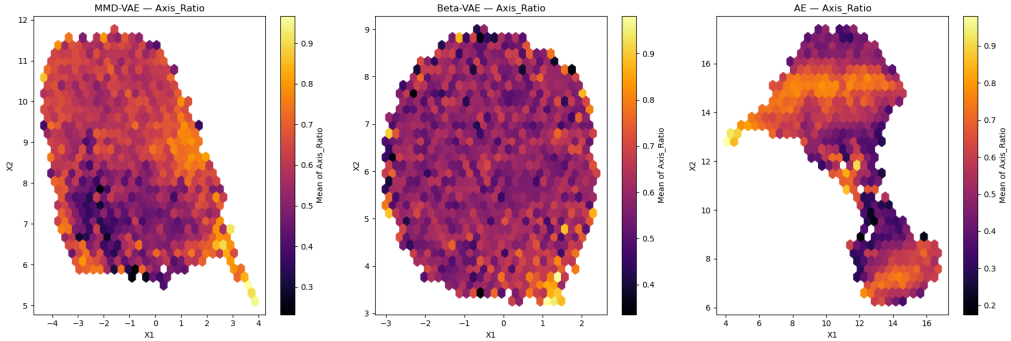
Analyzing the latent space

Latents correlate to physical parameters: asymmetry



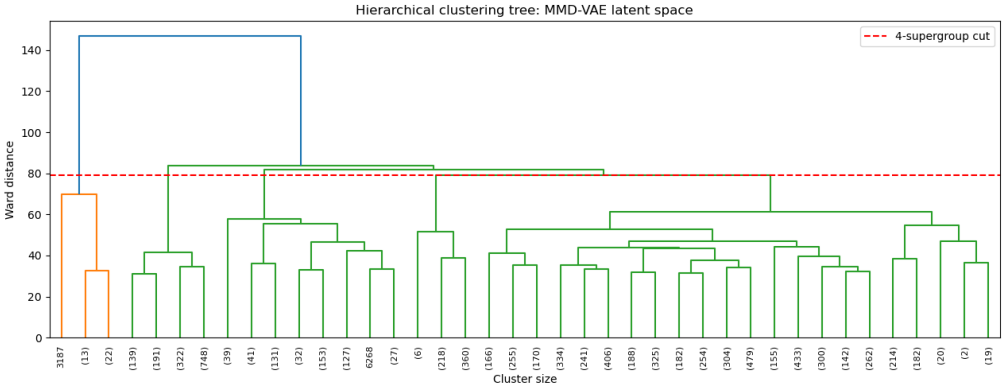
Analyzing the latent space

Latents correlate to physical parameters: axis ratio



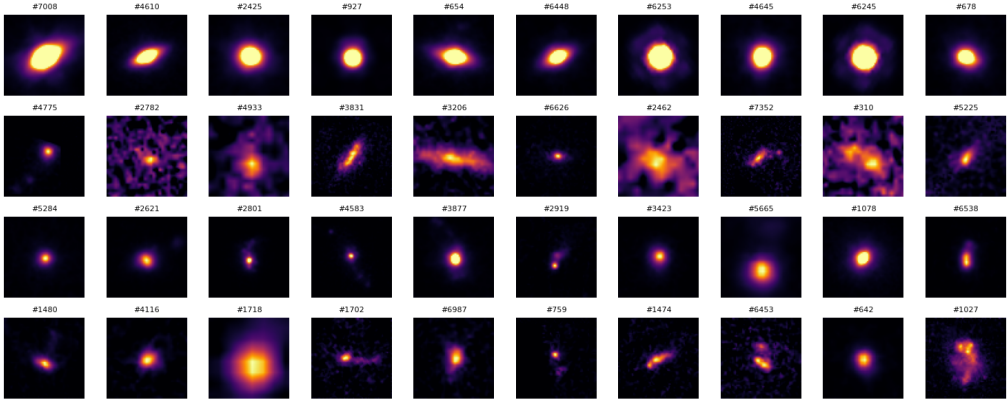
Clustering on MMD-VAE latents

We apply Hierarchical clustering to extract 4 classes



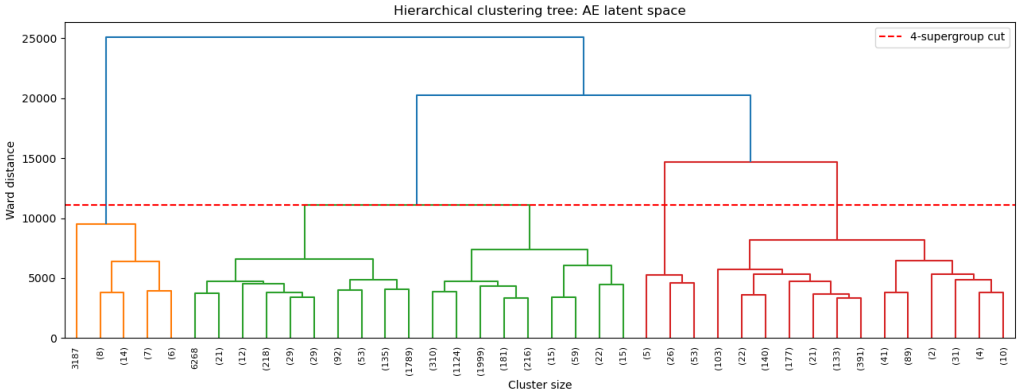
Clustering on MMD-VAE latents

Sample of the obtained classes



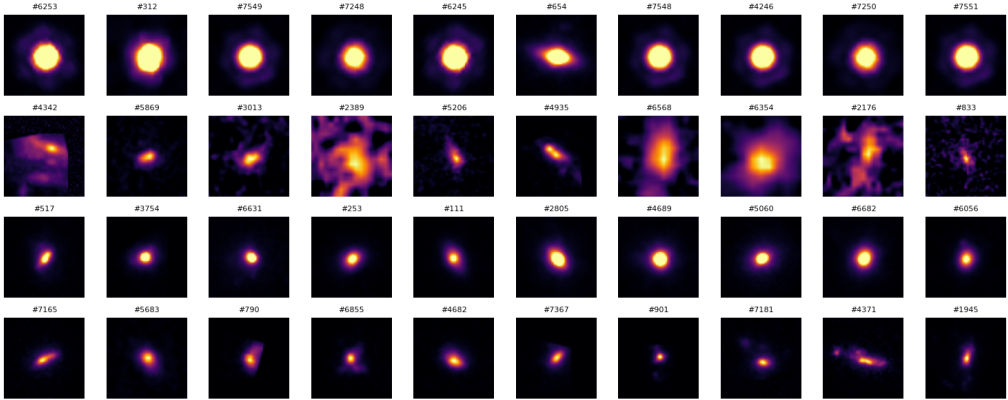
Clustering on AE latents

We apply Hierarchical clustering to extract 4 classes



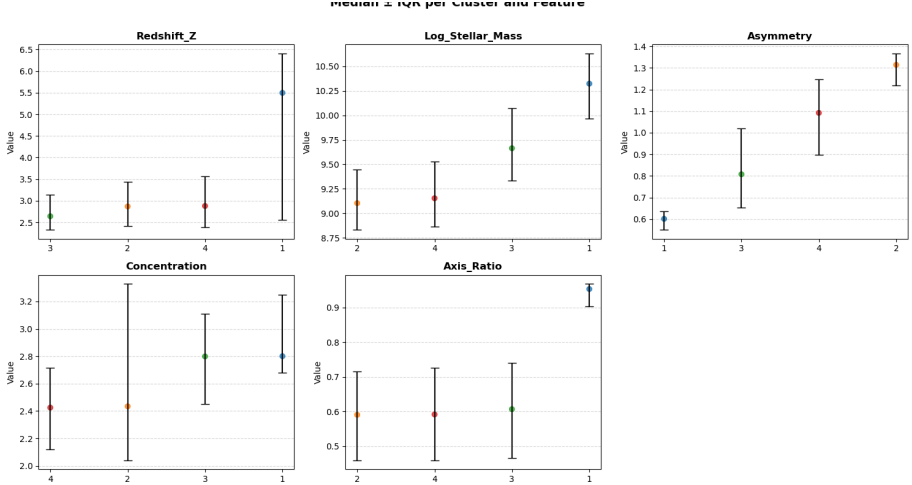
Clustering on AE latents

Sample of the obtained classes



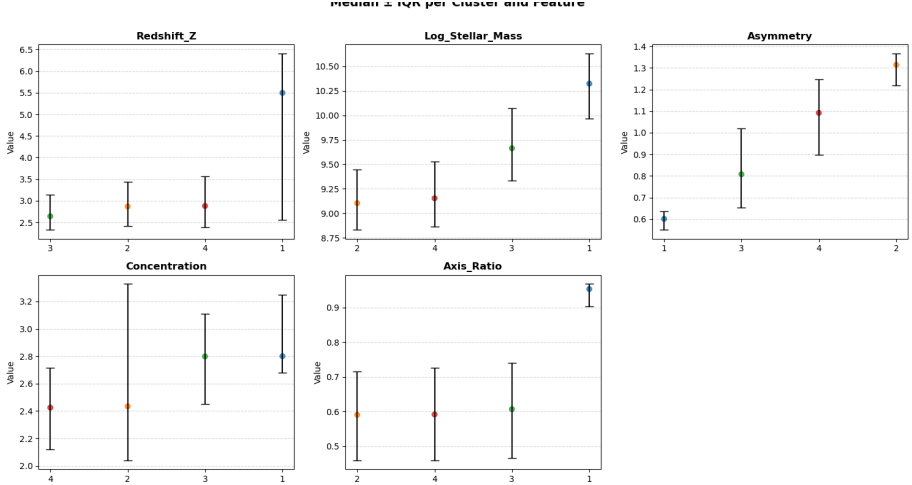
Clustering on MMD-VAE latents

Classes correlate to physical parameters

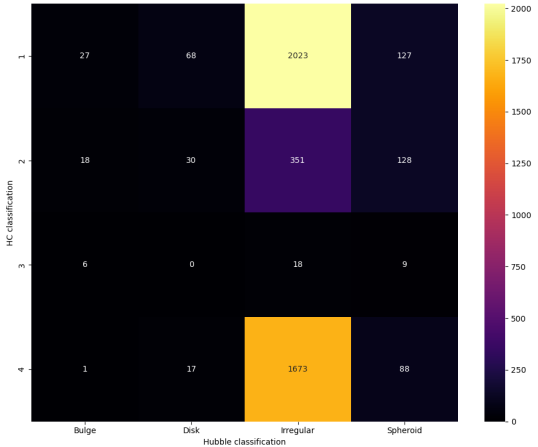


Clustering on AE latents

Surprisingly, simple AE works very well

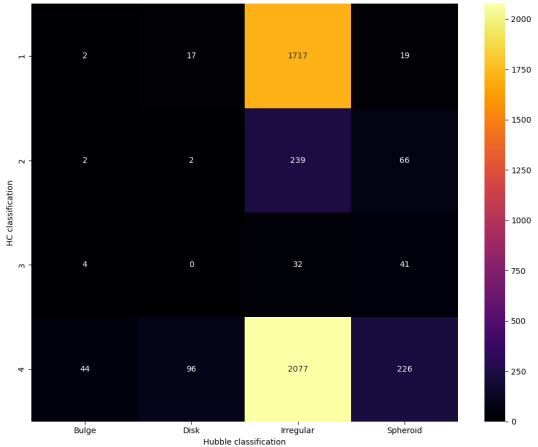


Compare with Hubble: MMD-VAE



Big class imbalance in Hubble labels

Compare with Hubble: AE



Big class imbalance in Hubble labels

Acknowledgements & references

- **Paper replicated:** Tohill C., et al., 2023, "Unsupervised Galaxy Morphological Classification at High-z with JWST/CEERS", `arXiv:2306.17225`.
- **Datasets:** CEERS DR1.0 (Cox+ 2025); CANDELS Santini+ 2015 mass catalog; COSMOS-Web DR1 (Shuntov+ 2025); Huertas-Company+ 2024 morphology catalog (VizieR `J/A+A/685/A48`).
- **Software:** `astropy`, `astroquery`, `photutils`, `scikit-image`, `pytorch`, `scikit-learn`, `umap-learn`, `seaborn`.
- **Image cleaning:** `galclean` reimplemented from de Albernaz Ferreira & Ferrari (2018).

Thank you!

Questions?

Antonio · Maria · Luxin